❶ Name of the node

❷ Nodes 1 and 2 are unreachable.

Here you can see the problem better. *Unreachability* is a relative term. When this cluster splits, its members remain in the Up state. So, if you have sharding entities, they are replicated on both sides of the cluster. Entities are missing on both sides, and the underlying mechanism restores them.

You have two clusters of actors that are no longer unique, at least when you consider the two clusters together. You can use a port connected to one side and change the value of one actor, and then, in the next call, use another port pointing to the other side and change a value in its duplicate. For example, suppose that wallet A is duplicated after a cluster split. If you deposit $100 on one side of the cluster and another $100 on the other, neither of the two wallets has received $200.

If your sharding entities are persisted, the problem gets worse. You can have two entities writing a different story in the same journal. Replaying those events would most likely be problematic.

To solve this problem, you need a strategy to shut down the nodes if one of them splits from the cluster. The only question is on which side. This is a hard problem, and the SBR helps you solve it, but it is not a panacea. There are corner cases where no solution is possible and the SBR shuts down all the nodes to end the cluster.

Before a node considers another node unreachable, a certain amount of time must pass during which the cluster view is stable: by default, `akka.cluster.split-brain-resolver.stable-`